

DEERWOOD, MB, Canada

WMO: 718400

Lat: 49.3997N

Lon: 98.3192W

Elev: 1120

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.8	-15.0	-25.6	1.4	-18.6	-21.6	1.7	-14.6	31.8	8.5	29.2	8.9	11.6	280	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.3	70.2	83.3	68.8	80.4	66.7	74.2	82.2	71.7	80.0	69.4	77.2	11.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	121.4	79.2	68.8	110.5	76.9	66.5	101.6	74.5	38.5	82.5	36.2	80.0	34.2	77.5	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.1	24.4	21.9	-25.1	93.4	3.7	3.1	-27.8	95.6	-30.0	97.5	-32.1	99.2	-34.8	101.5		
			-25.2	77.9	3.9	3.1	-28.0	80.2	-30.3	82.0	-32.5	83.7	-35.3	86.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	8.9	8.7	22.8	39.3	51.8	62.6	67.7	66.2	57.3	42.7	26.9	10.7
	DBStd	24.16	14.56	12.98	13.10	11.07	9.06	6.42	5.66	6.03	8.41	9.33	12.52	13.35
	HDD50	5925	1275	1155	846	348	88	2	0	0	28	268	696	1219
	HDD65	9804	1740	1575	1309	772	419	117	38	59	255	693	1143	1684
	CDD50	1891	0	0	2	27	143	380	548	501	247	40	3	0
	CDD65	297	0	0	0	0	10	45	121	95	25	1	0	0
	CDH74	2782	0	0	0	5	143	427	1006	916	272	13	0	0
	CDH80	707	0	0	0	0	33	86	241	279	65	3	0	0
	WSAvg	10.8	12.4	11.8	11.3	11.2	11.7	9.7	8.7	8.9	9.7	11.1	11.5	11.4
(15) Precipitation	PrecAvg	21.1	0.6	0.6	0.9	1.1	3.1	3.8	3.1	2.6	1.9	1.7	1.1	1.0
	PrecMax	26.9	1.4	2.1	2.5	2.2	6.5	7.0	7.8	5.6	4.1	6.1	3.4	2.2
	PrecMin	16.2	0.0	0.1	0.0	0.2	1.2	1.2	0.6	0.9	0.2	0.3	0.1	0.1
	PrecStd	3.1	0.3	0.5	0.6	0.6	1.5	1.5	1.5	1.3	0.8	1.2	0.8	0.5
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	41.5	61.5	74.0	84.5	86.3	89.3	91.4	86.5	74.8	62.1	41.3
		MCWB	37.6	36.6	51.8	55.6	60.8	71.9	75.2	72.3	65.4	59.4	52.1	36.6
	2%	DB	37.1	37.0	51.5	67.5	78.0	82.3	85.4	86.6	81.0	67.6	53.6	36.8
		MCWB	34.0	33.7	45.0	51.2	57.2	66.4	72.7	69.7	64.4	54.3	44.4	33.2
	5%	DB	33.7	32.9	43.7	62.9	72.4	78.8	82.6	83.0	76.8	62.0	48.3	33.1
		MCWB	31.4	30.9	39.2	48.7	55.8	64.9	71.0	68.6	62.8	51.7	42.0	30.7
	10%	DB	30.1	28.5	38.8	57.6	67.7	75.6	79.8	79.1	71.7	57.2	43.5	29.3
		MCWB	28.4	26.6	35.5	46.4	54.0	63.5	68.8	66.1	59.8	49.6	38.9	27.8
	0.4%	WB	37.4	37.0	53.2	58.4	64.8	75.1	78.4	76.8	71.1	61.1	51.5	37.2
		MCDB	41.2	41.0	58.2	67.4	77.3	82.3	85.5	85.3	81.6	71.7	61.8	41.6
	2%	WB	34.3	34.1	45.8	52.2	60.7	70.9	75.1	72.7	67.0	56.8	45.9	33.4
		MCDB	36.9	36.6	52.4	64.0	69.4	77.9	82.2	82.4	75.1	64.8	52.0	36.4
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	31.4	31.2	40.1	49.5	58.2	67.5	72.7	70.2	64.3	53.1	42.5	30.8
		MCDB	33.5	33.4	44.5	60.5	68.0	75.1	80.5	79.2	72.8	60.6	47.8	32.9
	10%	WB	27.7	27.0	36.3	46.8	55.9	64.9	70.4	67.9	61.7	50.0	39.0	27.4
		MCDB	29.5	28.8	39.1	56.5	65.6	72.7	77.8	76.3	69.8	57.2	43.4	28.9
	5% DB	MDBR	14.7	15.4	15.3	20.0	20.9	19.0	19.8	21.4	20.7	16.9	14.0	13.3
		MCDBR	17.2	17.1	19.3	28.6	28.3	24.2	22.5	26.4	27.3	24.6	20.9	17.1
(36) Mean Daily Temperature Range	5% WB	MCWBR	15.7	14.1	14.3	16.3	13.9	12.9	13.1	13.0	13.8	14.0	14.2	14.4
		MCDBR	17.5	16.4	18.8	26.4	22.3	21.0	21.4	23.1	24.1	22.7	19.8	16.3
	5% WB	MCWBR	15.7	14.0	14.1	16.0	13.2	13.2	13.8	13.3	14.4	13.9	13.9	14.3
		MCDBR	17.5	16.4	18.8	26.4	22.3	21.0	21.4	23.1	24.1	22.7	19.8	16.3
(40) Clear-Sky Solar Irradiance	taub		0.270	0.285	0.297	0.357	0.373	0.378	0.385	0.382	0.347	0.299	0.268	0.250
	taud		2.259	2.239	2.381	2.321	2.338	2.355	2.345	2.344	2.431	2.541	2.482	2.382
	Ebn at Noon		247	272	292	282	281	279	275	271	270	269	254	245
	Edh at Noon		23	30	31	37	38	38	37	36	30	22	18	17
(44) All-Sky Solar Radiation	RadAvg		470	847	1244	1583	1756	1891	2007	1648	1202	734	415	334
	RadStd		40	66	102	114	136	129	85	109	88	95	57	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page